MIL-C-11015/20C
28 September 1970
SUPERSEDING
MIL-C-11015/20B
6 December 1967

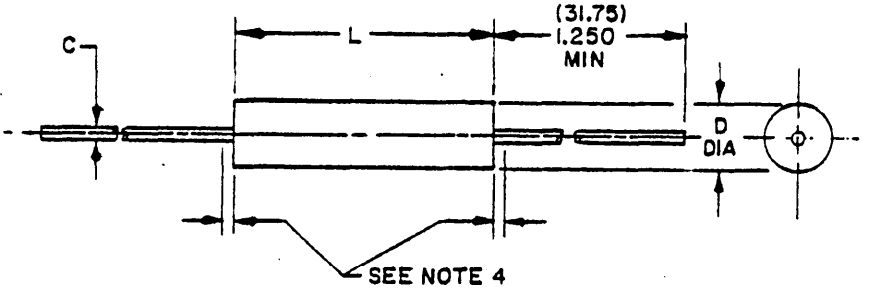
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE), STYLES CK12, CK13, CK14, CK15, AND CK16

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-11015.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

INACTIVE FOR NEW DESIGN
after 28 September 1970 use MIL-C-39014/5



Style	Dimensions					
	L		D		C	
	Inches	MM	Inches	MM	Inches	MM
CK12	.160±.010	4.06±.25	.090±.010	2.29±.25	.016+.004 −.001	.41+.10 −.03
CK13	.250±.010	6.35±.25	.090±.010	2.29±.25	.016+.004 −.001	.41+.10 −.03
CK14	.390±.010	9.91±.25	.140±.010	3.56±.25	.025+.004 −.001	.64+.10 −.03
CK15	.500±.020	12.70±.51	.250±.015	6.35±.38	.025+.004 −.001	.64+.10 −.03
CK16	.690±.030	17.53±.76	.350±.020	8.89±.51	.025+.004 −.001	.64+.10 −.03

NOTES:
1. All dimensions in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Non-conductive material shall not extend beyond .010 (.25 mm) from end of capacitor body.

FSC 5910

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(See instructions — Reverse)

TABLE I. Styles CK12, CK13, CK14, CK15, and CK16 characteristics.

Type designation 1/	Rated voltage	Rated temperature and voltage-temperature limits	Capacitance	Capacitance tolerance
	volts, dc		pF	
CK12BX100-	100	BX	10	K, M
CK12BX120K	100	BX	12	K
CK12BX150-	100	BX	15	K, M
CK12BX180K	100	BX	18	K
CK12BX220-	100	BX	22	K, M
CK12BX270K	100	BX	27	K
CK12BX330-	100	BX	33	K, M
CK12BX390K	100	BX	39	K
CK12BX470-	100	BX	47	K, M
CK12BX560K	100	BX	56	K
CK12BX680-	100	BX	68	K, M
CK12BX820K	100	BX	82	K
CK12BX101-	100	BX	100	K, M
CK12BX121K	100	BX	120	K
CK12BX151-	100	BX	150	K, M
CK12BX181K	100	BX	180	K
CK12BX221-	100	BX	220	K, M
CK12BX271K	100	BX	270	K
CK12BX331-	100	BX	330	K, M
CK12BX391K	100	BX	390	K
CK12BX471-	100	BX	470	K, M
CK12BX561K	100	BX	560	K
CK12BX681-	100	BX	680	K, M
CK12BX821K	100	BX	820	K
CK12BX102-	100	BX	1,000	K, M
CK12BX122K	100	BX	1,200	K
CK12BX152-	100	BX	1,500	K, M
CK12BX182K	100	BX	1,800	K
CK12BX222-	100	BX	2,200	K, M
CK12BX272K	100	BX	2,700	K
CK12BX332-	100	BX	3,300	K, M
CK12BX392K	100	BX	3,900	K
CK12BX472-	100	BX	4,700	K, M
CK12BX562K	50	BX	5,600	K
CK12BX682-	50	BX	6,800	K, M
CK12BX822K	50	BX	8,200	K
CK12BX103-	50	BX	10,000	K, M
CK13BX562K	100	BX	5,600	K
CK13BX682-	100	BX	6,800	K, M
CK13BX822K	100	BX	8,200	K
CK13BX103-	100	BX	10,000	K, M
CK13BX123K	50	BX	12,000	K
CK13BX153-	50	BX	15,000	K, M
CK13BX183K	50	BX	18,000	K
CK13BX223-	50	BX	22,000	K, M
CK13BR273K	50	BR	27,000	K
CK13BR333-	50	BR	33,000	K, M
CK13BR393K	50	BR	39,000	K
CK13BR473-	50	BR	47,000	K, M

See footnote at end of table.

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TABLE I. Styles CK12, CK13, CK14, CK15, and CK16 characteristics (cont).

Type designation 1/	Rated voltage	Rated temperature and voltage-temperature limits	Capacitance	Capacitance tolerance
	volts, dc		pF	
CK14BX123K	100	BX	12,000	K
CK14BX153-	100	BX	15,000	K, M
CK14BX183K	100	BX	18,000	K
CK14BX223-	100	BX	22,000	K, M
CK14BX273K	100	BX	27,000	K
CK14BX333-	100	BX	33,000	K, M
CK14BX393K	100	BX	39,000	K
CK14BX473-	100	BX	47,000	K, M
CK14BR563K	100	BR	56,000	K
CK14BR683-	100	BR	68,000	K, M
CK14BR823K	100	BR	82,000	K
CK14BR104-	100	BR	100,000	K, M
CK14BR124K	50	BR	120,000	K
CK14BR154-	50	BR	150,000	K, M
CK14BR184K	50	BR	180,000	K
CK14BR224-	50	BR	220,000	K, M
CK14BR274K	50	BR	270,000	K
CK15BX104-	100	BX	100,000	K, M
CK15BR124K	100	BR	120,000	K
CK15BR154-	100	BR	150,000	K, M
CK15BR184K	100	BR	180,000	K
CK15BR224-	100	BR	220,000	K, M
CK15BR274K	100	BR	270,000	K
CK15BR334-	100	BR	330,000	K, M
CK15BR474-	50	BR	470,000	K, M
CK15BR105-	50	BR	1,000,000	K, M
CK16BR474-	100	BR	470,000	K, M
CK16BR105-	100	BR	1,000,000	K, M
CK16BR225-	50	BR	2,200,000	K, M
CK16BR335-	50	BR	3,300,000	K, M

1/ Where applicable, the complete type designation will include an additional symbol to indicate capacitance tolerance.

REQUIREMENTS:

Design and construction:
Dimensions and configuration - See figure.
Case type - Tubular with axial leads.
Capacitance value - See table I.
Capacitance tolerance - ±10 percent (K) or ±20 percent (M) as shown in table I.
Rated temperature - -55° to +125° C.

Dielectric withstanding voltage:
Dielectric (DWV): Method 301 of MIL-STD-202.
Test voltage - 250 percent of rated voltage.
Body insulation: Not applicable.

Barometric pressure (reduced): Method 105 of MIL-STD-202. 0.82 inch of mercury (80,000 ft).
Test potential - 150 percent of rated voltage.

Insulation resistance (IR): Method 302 of MIL-STD-202, rated voltage applied. 100,000 megohms, min or 1,000 megohm-microfarads, min, whichever is less.

Capacitance (Cap.): Method 305 of MIL-STD-202.
Test frequency:
100 pF or less - 1 MHz ± 1 kHz.
Over 100 pF - 1 kHz ± 100 Hz.
Test voltage - 1.0 ± 0.2 volt rms.

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Dissipation factor (DF): 2.5 percent, max.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I (100 G's).

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20 G's).

Temperature cycling and immersion:
Temperature cycling: Method 102 of MIL-STD-202, condition D, except step 3 shall be at high test temperature.
Immersion: Method 104 of MIL-STD-202, condition B.

DWV - 250 percent of rated voltage.
IR - Not less than 50 percent of initial requirement.
Cap. - Within tolerance of table I value.
DF - 2.5 percent, max.

Salt spray (corrosion): Not applicable.

Terminal strength: In accordance with MIL-C-11015.
CK12 and CK13 - 3 pounds.
CK14, CK15, and CK16 - 5 pounds.

Moisture resistance: Method 106 of MIL-STD-202.
DWV - 250 percent of rated voltage.
IR - Not less than 50 percent of initial requirement.
Cap. - Within tolerance of table I value.

Solderability: Method 208 of MIL-STD-202 2 terminals.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B.
IR - 100,000 megohms, min or 1,000 megohm-microfarads, min, whichever is less.
ΔCap. - ±5 percent of initial measurement.
ΔDF - 0.5 percent, max.

Voltage temperature limits: In accordance with MIL-C-11015.

Life (at elevated ambient temperature):
Method 108 of MIL-STD-202. 2,000 hours (for qualification); 250 hours (performance check); 1,750 hours (continuation test).
Test potential - 200 percent of rated voltage.
IR - 5,000 megohms, min, or 50 megohm-microfarads, min, whichever is less (at 125° C) and not less than 50 percent of initial requirement (at 25° C).
DF - 3 percent, max (at 25° C).

Marking: In accordance with MIL-C-11015.

Inactivated styles: For new design it is recommended that units of MIL-C-39014/5 be considered as listed in table II. Although table II lists only the highest failure rate level, lower failure rate levels may be considered. For substitutability of the inactivated styles, capacitors in MIL-C-39014/5 of the same capacitance value, tolerance, rated voltage and temperature coefficient can be used regardless of the failure rate designation.

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TABLE II. Cross reference of inactivated styles.

From inactivated MIL-C-11015	To MIL-C-39014/5 L (2.0) M39014/05-	From inactivated MIL-C-11015	To MIL-C-39014/5 L (2.0) M39014/05-
CK12BX100K	0401	CK12BX682M	0453
CK12BX100M	0402	CK12BX822K	0454
CK12BX120K	0403	CK12BX103K	0455
CK12BX150K	0404	CK12BX103M	0456
CK12BX150M	0405	CK13BX562K	0457
CK12BX180K	0406	CK13BX682K	0458
CK12BX220K	0407	CK13BX682M	0459
CK12BX220M	0408	CK13BX822K	0460
CK12BX270K	0409	CK13BX103K	0461
CK12BX330K	0410	CK13BX103M	0462
CK12BX330M	0411	CK13BX123K	0463
CK12BX390K	0412	CK13BX153K	0464
CK12BX470K	0413	CK13BX153M	0465
CK12BX470M	0414	CK13BX183K	0466
CK12BX560K	0415	CK13BX223K	0467
CK12BX680K	0416	CK13BX223M	0468
CK12BX680M	0417	CK13BR273K	0469
CK12BX820K	0418	CK13BR333K	0470
CK12BX101K	0419	CK13BR333M	0471
CK12BX101M	0420	CK13BR393K	0472
CK12BX121K	0421	CK13BR473K	0473
CK12BX151K	0422	CK13BR473M	0474
CK12BX151M	0423	CK14BX123K	0475
CK12BX181K	0424	CK14BX153K	0476
CK12BX221K	0425	CK14BX153M	0477
CK12BX221M	0426	CK14BX183K	0478
CK12BX271K	0427	CK14BX223K	0479
CK12BX331K	0428	CK14BX223M	0480
CK12BX331M	0429	CK14BX273K	0481
CK12BX391K	0430	CK14BX333K	0482
CK12BX471K	0431	CK14BX333M	0483
CK12BX471M	0432	CK14BX393K	0484
CK12BX561K	0433	CK14BX473K	0485
CK12BX581K	0434	CK14BX473M	0486
CK12BX681M	0435	CK14BR563K	0493
CK12BX821K	0436	CK14BR683K	0494
CK12BX102K	0437	CK14BR683M	0495
CK12BX102M	0438	CK14BR823K	0496
CK12BX122K	0439	CK14BR104K	0497
CK12BX152K	0440	CK14BR104M	0498
CK12BX152M	0441	CK14BR124K	0499
CK12BX182K	0442	CK14BR154K	0500
CK12BX222K	0443	CK14BR154M	0501
CK12BX222M	0444	CK14BR184K	0502
CK12BX272K	0445	CK14BR224K	0503
CK12BX332K	0446	CK14BR224M	0504
CK12BX332M	0447	CK14BR274K	0505
CK12BX392K	0448	CK15BX104K	0510
CK12BX472K	0449	CK15BX104M	0511
CK12BX472M	0450	CK15BR124K	0512
CK12BX562K	0451	CK15BR154K	0513
CK12BX682K	0452	CK15BR154M	0514

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TABLE II. Cross reference of inactivated styles - Continued.

From inactivated MIL-C-11015	To MIL-C-39014/5 L (2.0) M39014/05-	From inactivated MIL-C-11015	To MIL-C-39014/5 L (2.0) M39014/05-
CK15BR184K	0515	CK15BR105M	0526
CK15BR224K	0516	CK16BR474K	0527
CK15BR224M	0517	CK16BR474M	0528
CK15BR274K	0518	CK16BR105K	0531
CK15BR334K	0519	CK16BR105M	0532
CK15BR334M	0520	CK16BR225K	0533
CK15BR474K	0521	CK16BR225M	0534
CK15BR474M	0522	CK16BR335K	0535
CK15BR105K	0525	CK16BR335M	0536

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
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Navy - AS, OS, SH
Air Force - 11, 17, 80
DSA - ES

User activities:
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Navy - MC
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

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